ASTRONOMY OLYMPIAD 2026

JUNIOR CATEGORY — PRACTICE PAPER

Unofficial preparation material • Created for workshop practice, not an official competition paper

Suggested duration: 35 minutes **Total:** 30 marks

Instructions
• Answer all questions.
• Section A: 15 multiple-choice questions, 1 mark each (15 marks).
• Section B: 5 short-answer questions, 3 marks each (15 marks).
• Show working for numerical questions. Award partial credit for correct reasoning.
• For this practice paper, there is no negative marking.

Section A — Multiple Choice (15 marks)

1. Which object is at the centre of our Solar System?
A. Earth B. The Sun C. The Moon D. Jupiter

2. What causes day and night on Earth?
A. Earth's revolution around the Sun B. The Moon's orbit C. Earth's rotation D. Clouds covering the Sun

3. Why do we have seasons?
A. Earth's tilt as it orbits the Sun B. Earth moves much closer to the Sun in summer C. The Moon blocks sunlight D. The Sun changes size each season

4. Which is the largest planet in our Solar System?
A. Earth B. Mars C. Saturn D. Jupiter

5. Which planet is famous for its prominent ring system?
A. Mercury B. Saturn C. Venus D. Mars

6. The Moon shines mainly because it…
A. makes its own visible light B. reflects sunlight C. is made of fire D. reflects light from Earth only

7. What is a star?
A. A rocky planet B. A natural satellite C. A hot ball of gas producing energy D. A comet

8. Why do stars appear to twinkle?
A. They switch on and off B. Earth's atmosphere bends their light C. They orbit Earth every second D. They are smaller than planets

9. Which object is made mostly of ice, dust, and rock and can develop a tail near the Sun?
A. Comet B. Asteroid only C. Moon D. Artificial satellite

10. What does a satellite do?
A. Always travels to another star B. Orbits a planet or another body C. Produces all its own light D. Never moves

11. Which layer of the Sun do we normally see as its visible surface?
A. Core B. Photosphere C. Corona only D. Radiative zone

12. What is a lunar eclipse?
A. The Moon passes between Earth and Sun B. Earth's shadow falls on the Moon C. The Sun passes behind Mars D. The Moon disappears permanently

13. Which galaxy contains our Solar System?
A. Andromeda B. Whirlpool C. Milky Way D. Sombrero

14. What is a telescope mainly used for?
A. Making distant objects easier to observe B. Changing a planet's orbit C. Creating stars D. Stopping Earth's rotation

15. What is an exoplanet?
A. A planet outside our Solar System B. A moon of Earth C. A type of galaxy D. A space telescope

Section B — Short Answer & Simple Problems (15 marks)

16. Explain in one or two sentences why the Moon has phases.

Answer: _____

_____

17. Name the four inner planets in order from the Sun.

Answer: _____

_____

18. Give one difference between a star and a planet.

Answer: _____

_____

19. What is one important use of artificial satellites?

Answer: _____

_____

20. A planet is 3 times as far from a lamp as another planet. If both receive light from the same lamp, which receives less light and why? (Use inverse-square idea.)

Answer: _____

_____

JUNIOR CATEGORY — ANSWER KEY & EXPLANATIONS

Use this section after students have completed the paper.

Section A — Multiple Choice

1. **B.** The planets orbit the Sun.
2. **C.** Earth rotates once roughly every 24 hours.
3. **A.** Earth's axial tilt changes the angle and duration of sunlight.
4. **D.** Jupiter is the largest planet.
5. **B.** Saturn's rings are especially prominent.
6. **B.** The Moon reflects sunlight.
7. **C.** Stars produce energy, mainly through nuclear fusion.
8. **B.** Atmospheric turbulence changes the apparent brightness and position.
9. **A.** Solar heating releases gas and dust from a comet.
10. **B.** Natural and artificial satellites orbit larger bodies.
11. **B.** The photosphere is the visible surface.
12. **B.** A lunar eclipse occurs when the Moon enters Earth's shadow.
13. **C.** Our Solar System is in the Milky Way.
14. **A.** Telescopes collect light or other electromagnetic radiation.
15. **A.** Exoplanets orbit stars beyond our Solar System.

Section B — Suggested Answers

16. The Moon orbits Earth, and we see different portions of its sunlit half from Earth.
17. Mercury, Venus, Earth, Mars.
18. A star produces energy and emits light; a planet usually shines by reflecting its star's light.
19. Examples include communication, weather forecasting, navigation, Earth observation, or scientific research.
20. The farther planet receives 1/9 as much light, because light intensity decreases with the square of distance.

Marking guidance: Each short-answer question is worth up to 3 marks. Give credit for scientifically correct equivalent wording. For numerical questions, award partial credit for a correct formula or method even if the final arithmetic is imperfect.